Informatics in Medicine Unlocked 41 (2023) 101326

Contents lists available at ScienceDirect

Informatics in Medicine Unlocked

journal homepage: www.elsevier.com/locate/imu

Systematic review of natural language processing for recurrent cancer detection from electronic medical records

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ARTICLE INFO

Keywords:
Recurrence
Cancer
Natural language processing
Clinical notes
Deep learning

ABSTRACT

This systematic review was conducted to explore natural language processing (NLP) focusing on text representation techniques and algorithms used previously to identify recurrent cancer diagnoses from electronic medical records (EMR), and an assessment of their detection performance. Relevant studies were identified from PubMed, Scopus, ACM Digital Library, and IEEE databases since inception to August 18, 2022. Data, including text representation methods, model algorithms and performance, and type of clinical notes, were extracted from individual studies by two independent reviewers. Study risk of bias was assessed using the prediction model risk of bias assessment tool. Of the 412 studies identified, 17 were eligible for inclusion, with 15 representing models that were not externally validated. Three text representations were used: statistical, context-free, and contextual representations (bidirectional encoder representations from transformers (BERT) and its variants), from 12, 6, and 3 studies, respectively. The corresponding median harmonic precision and recall means (F_1 scores) for these representations were 0.43, 0.87, and 0.72, respectively. The algorithms applied included rule-based, machine learning, and deep learning approaches with median F_1 scores of 0.71, 0.43, and 0.76, respectively. In conclusion, this systematic review suggests that deep learning models that use PubMedBERT as a text representation perform best. These findings are clinically informative for the selection of appropriate approaches for the detection of recurrent cancer from electronic medical records.

1. Introduction

Cancer remains a major public health concern, with 23.6 million cases and 10 million deaths reported globally in 2019 [1]. Many randomized control trials (RCT) [2] have evaluated therapies that delay disease progression and prolong patient survival. However, some patients experience cancer recurrence, which negatively impacts quality of life, often leading to premature mortality. Therefore, identification of recurrent cancer events represents an important outcome for the assessment of treatment efficacy and the effectiveness of RCTs and clinical studies. However, patient's characteristics for those enrolled in RCTs may differ to those from real world settings, therefore, evaluation of real-world data (RWD) provides clinically informative evidence. Such data includes electronic medical records (EMR), disease registries, and insurance claims. RWD reflects actual clinical practice and better represents the heterogeneity commonly observed among patients, in contrast to observations from RCTs [3]. However, RWD is often considered "messy", with multiple formats, sparse information, and incomplete treatment outcomes and prognoses. Furthermore, identifying recurrent cancer from RWD is time-consuming, labor-intensive,

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https://doi.org/10.1016/j.imu.2023.101326
Received 30 April 2023; Received in revised form 24 July 2023; Accepted 8 August 2023
Available online 10 August 2023
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and error-prone, resulting in limitations related to under-coding [4,5]. Due to these constraints, most hospitals may lack accurate cancer recurrence data due to under-detection, which limits the completeness of cancer registries at a population level [6].

Increasingly, digital technologies have been widely adopted in many healthcare settings to store RWD in electronic format. Structured data in standardized data formats such as diagnosis, laboratory, and medication may aid identification of cancer recurrence [7], some limitations cannot be avoided. More than 70%–80% of cancer diagnoses from RWD are collected in unstructured formats, such as clinical notes, diagnosis reports, discharge summaries, and radiology reports, which represent rich sources of information [8]. However, these data are not easily useable, requiring manual EMR review to extract recurrence information. To address these challenges, several automated tools have been developed to detect disease recurrence from clinical textual data using natural language processing (NLP) [9–12].

NLP is a computational approach to learn, understand and generate human language content [13]. Various NLP applications, such as machine translation, speech-to-text, text generation, and text classification [14], have been adopted in real-world practice to extract information from unstructured text data and convert it into structured or tabular formats [15,16].

NLP techniques with various text representation methods have been applied, including statistical text representation (such as bag-of-words and term frequency-inverse document frequency (TF-IDF) [17]), context-free word embeddings (such as word2vec [18], global vectors for word representation (GloVe) [19]), and contextual word embeddings (such as embeddings from language models (ELMo) [20], universal language model fine-tuning for text classification (ULMfit) [21], and bidirectional encoder representations from transformers (BERT) [22]). These techniques are applied in adjunct with model algorithms to predict events of interest such as cancer recurrence. These algorithms tend to be: 1) rule-based following a set of defined rules or patterns based on human knowledge to support NLP tasks. 2) ML methods that enable learning from patterns hidden within training data that can make predictions or decisions without explicit programming. Examples of ML algorithms that have been applied in recurrent cancer detection include support vector machine (SVM) [23] and logistic regression approaches (LR) [24]. 3) DL models consist of mechanisms to derive optimum solutions from complex and relevant input datasets [25]. DL algorithms such as convolutional neural networks (CNN) [26], recurrent neural networks (RNN) [27], long short-term memory (LSTM) [27], bidirectional-LSTM (BiLSTM) [28], autoencoders and transformers [29], utilize artificial neural networks (ANN) [30] with multiple layers to model complex patterns and data representations.

Several studies have previously applied these text representations and algorithms to automatically detect cancer recurrence from EMRs, focusing on specific study populations and clinical documents, resulting in varied detection performance. These approaches offer valuable insight for the development of recurrent cancer detection techniques in real-world settings. However, a systematic review of these approaches for recurrent cancer detection and assessment of their effectiveness, has yet to be undertaken.

Therefore, this systematic review was conducted with the following aims: Firstly, to explore and understand the current NLP methods for textual analysis in all types of cancer patients, including text representation techniques, and model algorithms. Secondly, to evaluate these ML and DL model performance for recurrent cancer prediction.

2. Material and methods

A systematic review was conducted in accordance with the preferred reporting items for systematic reviews and meta-analyses (PRISMA2020) statement [31], and the transparent reporting of multivariable prediction models for individual prognosis or diagnosis: checklist for systematic reviews and meta-analyses (TRIPOD-SRMA) [32]. Our review protocol was registered in PROSPERO (CRD42022370805, available at https://www.crd.york.ac.uk/prospero/display_record.php?ID=CRD42022370805l).

2.1. Search strategy

The search databases included Medline via PubMed, Scopus, ACM Digital Library, and IEEE Xplore Digital Library were used to identify relevant studies from the date of inception (or establishment) for each database to August 18, 2022. These databases were selected based on their relevance to medicine, bioscience, computer science, and technology. There were no language restrictions imposed provided publications were in English or amenable to Google Translate.

The search terms and strategies were constructed based on patients (cancers), input data (clinical text, clinical notes, electronic medical records, progress and admission notes, radiology and pathology reports, natural language processing, machine learning, and deep learning), and outcomes (recurrence, metastasis, and progression). Search strategy details and queries are shown in Supplementary Tables 1–2. Studies identified were imported into the EndNote software for removal of duplicate studies before the screening of titles, abstracts and full-papers.

2.2. Inclusion and exclusion criteria

Two reviewers (ES, WP) independently selected studies based on the following inclusion and exclusion criteria. Any disagreements were resolved with the assistance of a third reviewer (ATA).

2.2.1. Inclusion criteria

Studies were eligible if they met the following criteria: 1) included adults with any specific type of cancer or mixed cancers, 2) used any clinical document (such as clinical or progress notes, discharge summaries), 3) applied any NLP techniques, 4) used any rule-based, ML, or DL algorithm, and 5) classified any cancer as recurrence or metastasis. Text representations and algorithms used are described in Table 1.

2.2.2. Exclusion criteria

Studies were excluded if: 1) they focused on hematologic cancers, 2) were published in any non-English language not amenable to translation by Google Translate, 3) were limited to a published abstract without provision of complete data for extraction.

Table 1
Details of natural language processing text representations and algorithms included.

Text representations	Techniques
Statistical text representation	Concept match, TF-IDF, Bag-of-word, Name entity tag, Negation tag, Temporality tag, Clinical term
Context-free word embeddings	Word2Vec, CUI2Vec
Contextual word embeddings	BERT, ClinicalBioBERT, BioBERT, ClinicalBERT, BlueBERT, PubMedBERT, MetBERT
Algorithms	**Techniques**
Rule-based	The pattern or keyword matching. For instance, if the sentence presented a targeted clinical term without negation or historical context, it was defined as a definite recurrence.
ML	LR, SVM, SVC, RF, NB, SGD, XGBoost
DL	DNN, CNN, LSTM, BiLSTM, Linear classifying layer

Abbreviation: BERT: bidirectional encoder representations from transformers; BiLSTM: bidirectional long short-term memory; CNN: convolutional neural network; DL: deep learning, DNN: deep neural network; LR: logistic regression; LSTM: long short-term memory; ML: machine learning; NB: naïve Bayes; RF: random forest; SGD: stochastic gradient classifier; SVM: support vector machine; SVC: linear support vector classifier; XGBoost: extreme gradient boosting.

2.3. Data extraction

Data were independently extracted using a pre-specified extraction form (see Supplementary Table 3) by the same two reviewers. Any disagreements between reviewers were resolved through consensus with a third reviewer, as previously mentioned. Relevant data were extracted from individual studies, including study characteristics (authors, year), study phase (development or external validation), data source (local or public data), primary organ cancer (specific organs, including breast, lung, colorectal cancers, or mixed cancer types), number of patients, site of organ recurrence (specific or any organ), type of recurrence (local or distant recurrence), and clinical document data characteristics, including EMRs, nurse or progress notes, or pathology and radiology reports.

Development and external validation study phases were documented; development phase included training, validation, and internal testing of the model normally undertaken by splitting the data for training, validation, and test data to assess model performance. External validation on data from independent sources provides an evaluation of the applicability or generalizability of the model developed.

In addition, data on NLP text representation methods, the number of documents and words, classifiers/algorithms used (including rule-based, ML, and DL), and model performance, were also extracted.

2.4. Risk of bias assessment

Risk of bias (ROB) assesses the predictive accuracy of models to estimate their transparency, bias and applicability. ROB was assessed using the prediction model risk of bias assessment tool (PROBAST) [33], which considers 4 domains. First, the participant domain evaluates participant selection methods and data source or study design. Second, predictor definitions and measurements are considered for potential bias and applicability. Third, outcomes are assessed for study definitions and measurements that may be subject to bias and applicability. Fourth, statistical analyses are evaluated for choice of feature extraction or text representation and algorithm, feature selection and hyper parameterization. Each domain was evaluated for risk of bias as "yes", "probably yes", "probably no", "no" or "no information", in which "yes (+)" and "no (−)" indicate low and high bias, respectively. Finally, studies were judged as high ROB if at least one domain was considered high risk; low ROB if all domains were deemed low risk, otherwise ROB was considered unclear.

2.5. Ethical considerations

This systematic review used previously published secondary data and therefore did not require ethical approval.

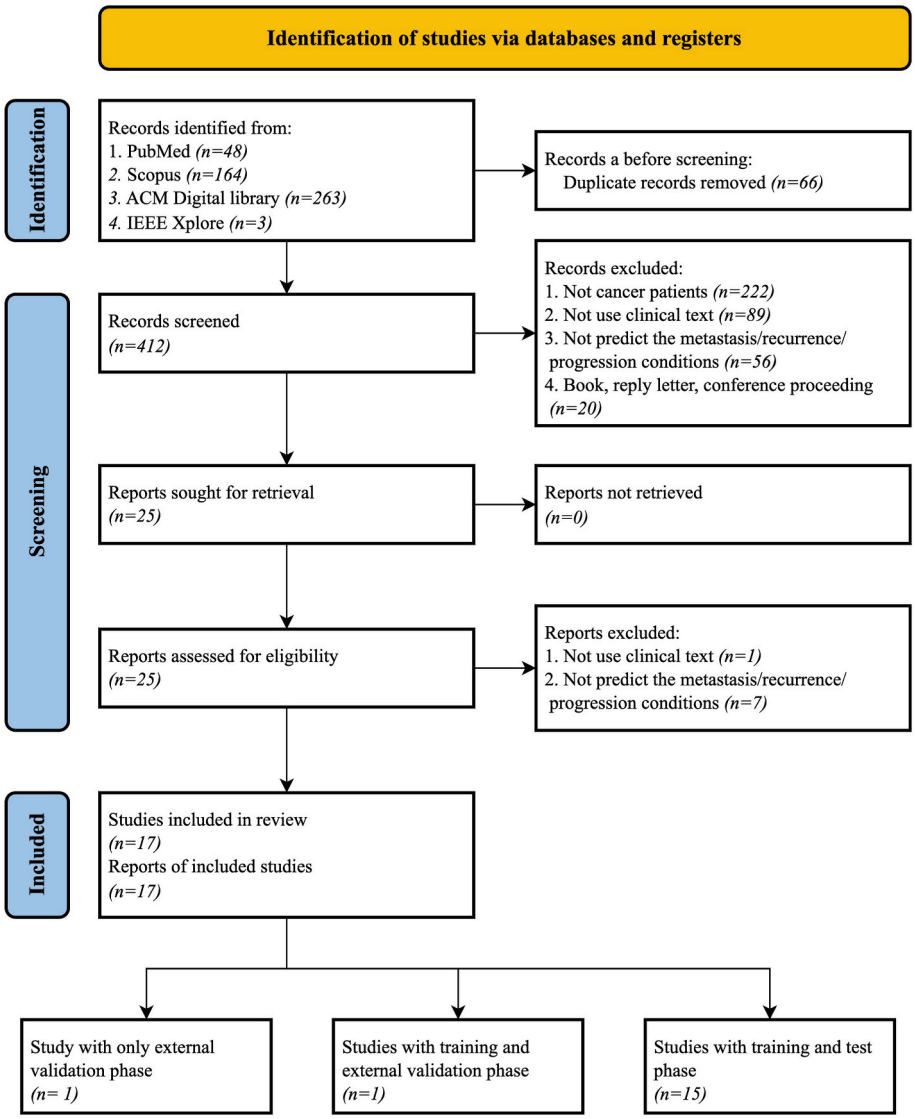

Fig. 1. PRISMA flow diagram for study selection.

3. Results

3.1. Selection of included studies

A total of 17 studies [10–12,34–47] from the 412 identified met the inclusion criteria (Fig. 1). Of these, 15 [10,11,34–46] focused only on a model development phase, a single study [47] included both development and external validation phases, and another study [12] was limited to only an external validation phase.

3.2. Characteristics of included studies

Of the 17 studies included, nine (53%) were recently published (2021 and 2022) [39–47], see Table 2. Most studies used local-data sources, while a single study [47] used public-data from the Medical Information Mart for Intensive Care (MIMIC-III) [48]. All studies applied NLPs to English-clinical notes, including pathology reports (N = 7), radiology reports (N = 10), and other clinical documents (N = 12).

Seven studies [12,40–43,46,47] included multiple cancer types in their study population. The three most common types included breast (N = 13), colorectal (N = 4), and lung cancers (N = 3). Outcomes of interest included metastasis/distant (N = 10), local (N = 2), and any recurrence (N = 6). Seventeen studies provided organ-specific recurrent cancers, including liver (N = 5), lymph nodes (N = 4), lung (n = 4), and bone (n = 4).

Patient numbers ranged from 220 to 22,543 with a median of 1,254

Table 2
Characteristics of the studies included in this review.

Study	Study-phase	Data source	Type of documents	No. of documents	Primary cancer	No. of patients	Outcomes of interest	Site of recurrence
Strauss J. (2012) [12]	Ext validation	Local	PR	915	Breast, prostate	206, 186	Any recurrence	Any organ
Carrell D. (2014) [11]	Development	Local	PR, RR, PN	241,652	Breast	1,472	Any recurrence	Any organ
Zeng Z. (2018) [10]	Development	Local	PN	6,899	Breast	490	Local recurrence	Local
Banerjee I. (2019) [34]	Development	Local	PR, RR, PN	173,264	Breast	894	Distant recurrence	Any organ
Ling A. (2019) [36]	Development	Local	Medical and social histories, visit summaries	NM	Breast	8,811	Distant recurrence	Any organ
Kehl K. (2019) [35]	Development	Local	RR	11,182	Lung	2,406	Distant recurrence	Liver, bone, brain, lymph nodes, adrenal
Zeng Z. (2019) [37]	Development	Local	PR, RR, PN, NN, other types of EMR	4,904	Breast	599	Distant recurrence	Any organ
Wang H. (2020) [38]	Development	Local	PR, PN	NM	Breast	6,447	Distant recurrence	Any organ
Alzu'bi A. (2021) [39]	Development	Local	EMR	142	Breast	142	Any recurrence	Any organ
Do R. (2021) [40]	Development	Local	RR	385,410	Colorectal, pancreas, stomach, esophagus	2,219	Any recurrence	Lung, pleura, lymph nodes, liver, spleen, adrenal, renal, pelvic organ, peritoneum, bowel, bones, soft tissue
Karimi Y. (2021) [41]	Development	Local	Clinician notes, PR, RR	NM	Breast, liver	1,075, 148	Local, distant recurrence	Any organ
Kehl K. (2021) [42]	Development	Local	RR	31,196	Breast, lung, colorectal, pancreas, prostate, kidney, bladder	2,830	Distant recurrence	Brain, bone, adrenal, lung, liver, lymph nodes, peritoneum
Kirschner J. (2021) [43]	Development	Local	Visit notes, RR	NM	Breast, lung, colorectal, prostate, kidney, bladder, melanoma	66,732	Any metastasis	Any organ
Sanyal J. (2021) [44]	Development	Local	PN, PR, RR, NN	894	Breast	224	Any recurrence	Any organ
Batch K. (2022) [45]	Development	Local	RR	14,617	NM	1,586	Any metastasis	Lung, liver, adrenal
Kaka H. (2022) [46]	Development	Local	EMR	429,162	Breast, colorectal	1,512, 1,125	Any recurrence	Any organ
Lui K. (2022) [47]	Development, Ext validation	Local and Public	Discharge summary	6,634	Any cancers	3,088	Distant recurrence	Any organ

Abbreviations: EMR: electronic medical record; NM: not mentioned; NN: nurse note; PN: progress note; PR: pathology report; RR: radiology report; Extvalidation: external validation.

(IQR:542-6,493) for the development phase and ranged between 47 and 6,447 with a median of 193 (IQR: 139-603) for the test data. The number of documents ranged from 1,557 to 127,981 with a median of 1,767 (IQR:1,740-4,144) for the development phase and between 142 and 241,652 with a median of 443 (IQR:426-1,417) for the test data, respectively. Most studies split data into training and test data with ratios varying between 90%:10% and 70%:30%.

For ROB assessment, 7, 9, and a single study were rated as low, unclear, and high ROB, respectively, see Table 3. The unclear and high ROB classifications were based on participant and outcome domains. Specifically, some studies failed to sufficiently clarify their respective eligibility criteria and/or the definition of recurrence outcomes.

3.3. Model development and test data performance

Four commonly reported evaluation metrics were considered: including the area under the receiver operating characteristic (AUROC), F_1, recall, precision, and specificity scores; the area under the precision and recall curve (AUPRC) is a harmonic mean of precision and recall but was rarely reported [35]. Model performance with regards to text representation and model used for studies included is summarized by cancer type in Table 4.

3.3.1. Text representation

A total of 267 models were considered from the 17 studies eligible for inclusion, given each individual study applied at least one or more models (see Fig. 2a). Of these, 130, 107, and 30 models were statistical text representation, context-free, or contextual word embeddings. The 130 models based on statistical text representations from 12 studies [10–12,34,36–41,43,45] used the following techniques: five applied TF-IDF [10,38–40,45], six applied concept matching [10–12,36–38], and three applied bag-of-words or clinical term methods [34,41,43]. Their median AUROC, F_1, recall, precision, and specificity in the data tested were 0.84, 0.43, 0.50, 0.65, and 0.95 (see Fig. 3). The context-free word embedding approach was reported in six studies [35,37,38,42,44,45], that included 107 models, in which three studies [37,38,44] applied self-trained word2vec from the MIMIC-III data, and a single study [38] used pre-trained word embeddings (CUI2vec), and three studies [35,42,45] applied unknown pre-trained word embeddings. The median AUROC, F_1, recall, precision, and specificity were 0.98, 0.87, 0.89, 0.89, and 0.95, respectively. The contextual word embeddings approach (BERT and its variants) was reported in only three studies [40,46,47] with median AUROC, F_1, recall, precision, and specificity scores in the data tested of 0.98, 0.72, 0.81, 0.59, and 0.99, respectively.

Our findings suggest models based on context-free word embedding approaches perform best with AUROC (0.98), F_1 (0.87), recall (0.89), and precision (0.89), while contextual word embedding techniques provide slightly higher specificity (0.98 vs. 0.95).

3.3.2. Model algorithms

Of the 16 studies that developed multiple algorithms, 11, nine, and three studies (see Fig. 2b) included 147 DL, 116 ML, and four rule-based algorithms, respectively. Common ML models included LR (N = 5) [36,38–40,43], SVM (N = 4) [10,37,39,40], and ensemble models (N = 4) [38,40,44,45]. DL models included CNN (N = 5) [35,37,38,42,45], LSTM (N = 2) [44,45], vanilla neural networks (N = 2) [34,41], and linear classifier of BERT (N = 3) [40,46,47].

DL algorithms performed had median scores of 0.98, 0.76, 0.88, 0.79, and 0.97 for AUROC, F_1, recall, precision, and specificity, compared to corresponding values of 0.84, 0.43, 0.51, 0.67, and 0.95 for ML approaches, respectively (Fig. 4).

3.4. Model performance in external validation datasets

Two studies [12,47] conducted external validation for three algorithms. The first study [12] performed external validation using a rule-based algorithm, achieving a recall score of 0.97–1 and a specificity of 0.99. The second study only externally validated the MetBERT algorithm, which was originally trained on MIMIC-III data, on local data [47], reporting an AUROC score of 0.94.

4. Discussion

We performed a systematic review of 17 studies to explore text representations and algorithms previously reported to identify recurrent cancers from clinical text from EMRs. Most of the studies included were conducted post 2018, highlighting recent developments in advanced NLP techniques to identify cancer recurrence from EMRs. Since 2019, context-free and contextual word embedding approaches have been used more frequently, resulting in improved model performance. Furthermore, DL approaches have been shown to outperform ML and rule-based algorithms across all evaluation metrics.

Breast cancer was the most common cancer type reported, largely due to improved treatment options and efficacy resulting in prolonged patient survival [49]. Nevertheless, several studies used text data reporting for various cancers [12,41–43,46] for model development,

Table 3
Risk of bias assessment of the studies included.

Study	Risk of bias (ROB)				Applicability			Overall	
	Participants	Predictors	Outcome	Analysis	Participants	Predictors	Outcome	ROB	Applicability
Strauss J. (2012) [12]	?	?	+	?	?	+	+	?	?
Carrell D. (2014) [11]	+	+	+	+	+	+	+	+	+
Zeng Z. (2018) [10]	?	+	?	+	?	+	?	?	?
Banerjee I. (2019) [34]	?	+	?	+	?	+	?	?	?
Ling A. (2019) [36]	+	+	+	+	+	+	+	+	+
Kehl K. (2019) [35]	+	+	+	+	+	+	+	+	+
Zeng Z. (2019) [37]	+	+	+	+	+	+	+	+	+
Wang H. (2020) [38]	?	+	?	+	?	+	?	?	?
Alzu'bi A. (2021) [39]	?	+	?	+	?	+	?	?	?
Do R. (2021) [40]	+	+	+	+	+	+	+	+	+
Karimi Y. (2021) [41]	–	+	+	?	–	+	+	–	–
Kehl K. (2021) [42]	+	+	?	?	+	+	?	?	?
Kirschner J.(2021) [43]	+	+	+	?	+	+	+	?	+
Sanyal J. (2021) [44]	+	+	+	?	+	+	+	?	+
Batch K. (2022) [45]	+	+	+	+	+	+	+	+	+
Kaka H. (2022) [46]	+	+	+	+	+	+	?	+	?
Lui K. (2022) [47]	?	+	?	+	?	+	?	?	?

Abbreviations: + indicates low ROB/low concern regarding applicability; – indicates high ROB/high concern regarding applicability; and ? indicates unclear ROB/unclear concern regarding applicability.

Table 4
Describe text representation, algorithms, and model performance of the included studies stratified by cancer types.

Study	Text representation type	Algorithm type	Model performance [Median (IQR)]				
			AUROC	F1 score	Precision	Recall	Accuracy
Breast cancer							
Carrell D. (2014) [11]	Statistics: Concept match	Rule-based	–	0.76 (0.74–0.8)	0.66 (0.62–0.71)	0.92 (0.92–0.92)	–
Zeng Z. (2018) [10]	Statistics: Concept match, TF-IDF	ML: SVM	0.84 (0.8–0.88)	0.67 (0.51–0.75)	0.72 (0.68–0.8)	0.57 (0.43–0.76)	–
Banerjee I. (2019) [34]	Statistics: Name entity tag, negation tag, temporality tag	Rule-based	–	0.57 (0.51–0.64)	–	0.52 (0.47–0.58)	–
	Statistics: Clinical term, term frequency	DL: Deep neural network	–	0.94 (0.94–0.94)	–	0.93 (0.93–0.93)	–
Ling A. (2019) [36]	Statistics: Concept match, TF-IDF	ML: SVM	0.92 (0.92–0.92)	0.87 (0.87–0.87)	0.87 (0.87–0.87)	0.86 (0.86–0.86)	–
Zeng Z. (2019) [37]	Statistics: Concept match, Bag-of-words	ML: SVM, MLP	0.93 (0.74–0.94)	0.67 (0.52–0.76)	0.72 (0.59–0.78)	0.62 (0.5–0.73)	–
	Statistics: Concept match	DL: CNN	0.71 (0.53–0.9)	0.24 (0.04–0.44)	0.35 (0.09–0.61)	0.18 (0.03–0.36)	–
	Context-free: Word2Vec	DL: CNN	0.92 (0.91–0.93)	0.61 (0.56–0.67)	0.75 (0.73–0.77)	0.53 (0.46–0.61)	–
Wang H. (2020) [38]	Statistics: Concept match, TF-IDF	ML: LR, RF, linear SVC, NB, SGD	0.84 (0.82–0.86)	0.36 (0.27–0.41)	0.52 (0.35–0.71)	0.33 (0.16–0.47)	–
	Context-free: Word embedding, CUI2vec	DL: CNN	0.87 (0.84–0.88)	0.42 (0.41–0.45)	0.48 (0.48–0.54)	0.37 (0.31–0.47)	–
Alzu'bi A. (2021) [39]	Statistics: TF-IDF	ML: Algorithms from WEKA	–	–	–	0.9 (0.89–0.91)	–
Sanyal J. (2021) [44]	Context-free: Word2Vec	ML: XGBoost	0.79 (0.78–0.8)	–	–	0.71 (0.69–0.74)	–
	Context-free: Word2Vec	DL: LSTM	0.92 (0.88–0.93)	–	–	0.83 (0.77–0.86)	–
Breast cancer with other cancer type							
Strauss J. (2012) [12]	Statistics: Concept match	Rule-based	–	0.97 (0.97–0.97)	0.96 (0.95–0.96)	0.98 (0.98–0.99)	–
Karimi Y. (2021) [41]	Statistics: Clinical term, term frequency	DL: Deep neural network	0.96 (0.96–0.97)	–	–	0.89 (0.88–0.9)	0.87 (0.85–0.88)
Kehl K. (2021) [42]	Context-free: Word embedding	DL: CNN	0.99 (0.98–0.99)	0.85 (0.8–0.89)	0.84 (0.79–0.87)	0.9 (0.87–0.91)	–
Kirschner J. (2021) [43]	Statistics: Clinical term	ML: LR	–	0.87 (0.84–0.9)	0.91 (0.89–0.92)	0.84 (0.81–0.89)	–
Kaka H. (2022) [46]	Contextual: ClinicalBioBERT	DL: Linear classifying layer	0.99 (0.99–0.99)	0.69 (0.64–0.73)	0.57 (0.54–0.6)	0.87 (0.79–0.93)	–
Lung cancer							
Kehl K. (2019) [35]	Context-free: Word embedding	DL: CNN	0.96 (0.92–0.97)	0.77 (0.71–0.83)	–	–	–
Mixed cancer types							
Do R. (2021) [40]	Statistics: TF-IDF	ML: LR, RF, SVM, XGBoost	–	0.75 (0.64–0.81)	0.67 (0.56–0.8)	0.83 (0.32–0.91)	0.98 (0.96–0.99)
	Context: BERT	DL: Linear classifying layer	–	0.69 (0.43–0.79)	0.68 (0.43–0.8)	0.81 (0.14–0.89)	0.98 (0.95–0.99)
Batch K. (2022) [45]	Statistics: TF-IDF	ML: LR, RF, SVM, XGBoost	–	0.84 (0.75–0.94)	0.94 (0.91–1.0)	0.78 (0.67–0.89)	0.96 (0.92–0.99)
	Context-free: Word2Vec	DL: CNN, BiLSTM	–	0.97 (0.92–0.98)	0.97 (0.94–0.99)	0.97 (0.9–0.99)	0.99 (0.98–1.0)
Lui K. (2022) [47]	Contextual: BERT, BlueBERT, BioBERT, ClinicalBERT, PubMedBERT, MetBERT	DL: Linear classifying layer	0.94 (0.94–0.94)	0.72 (0.67–0.74)	0.7 (0.58–0.72)	0.78 (0.72–0.78)	–

Abbreviations: BERT: bidirectional encoder representations from transformers; BiLSTM: bidirectional long short-term memory; CNN: convolutional neural network; IQR: interquartile range; LR: logistic regression; LSTM: long short-term memory; MLP: multilayer perceptron; NB: naïve Bayes; SGD: stochastic gradient descent; RF: random forest; SVM: support vector machines; TF-IDF: term frequency–inverse document frequency; XGBoost: extreme gradient boosting.

which improved performance and AUROC, F_1 scores (range from 0.96 to 0.99 and 0.69–0.97) compared to model performance from specific cancers [10,11,34,36–39,44] with AUROCs of 0.71–0.96 and F_1 scores of 0.24–0.97. Data related to different cancer types may increase model uncertainty given the greater data heterogeneity. However, this shortcoming could be addressed through the use of standardized medical terminology and document templates [50], especially for pathology [51] and radiology reports [52], a common source in most studies. Therefore, NLP techniques can be focused on specific or mixed cancer types if text normalizations have been applied to a particular setting.

The majority of studies that applied statistical text representations (such as TF-IDF [10,38,40,42,45] and concept matching [10–12, 36–38]) were published pre-2019. These techniques utilize bag-of-words models that create a numerical representation based on word occurrence. In addition, the matching method attempts to pair words within a sentence to clinical terms of similar meaning using CUI [53], such as mapping "bone metastases" to CUI code 0153690 (secondary malignant neoplasm of bone). This approach is intuitive and more easily understood but disregards the positioning of the words, different words with similar meaning, and dependencies between words within sentences. Context-free word embedding was developed [18] to improve consideration of word positioning within sentences.

Six studies [35,37,38,42,44,45] applied context-free word embedding using algorithms trained on healthcare data (CUI2vec) or

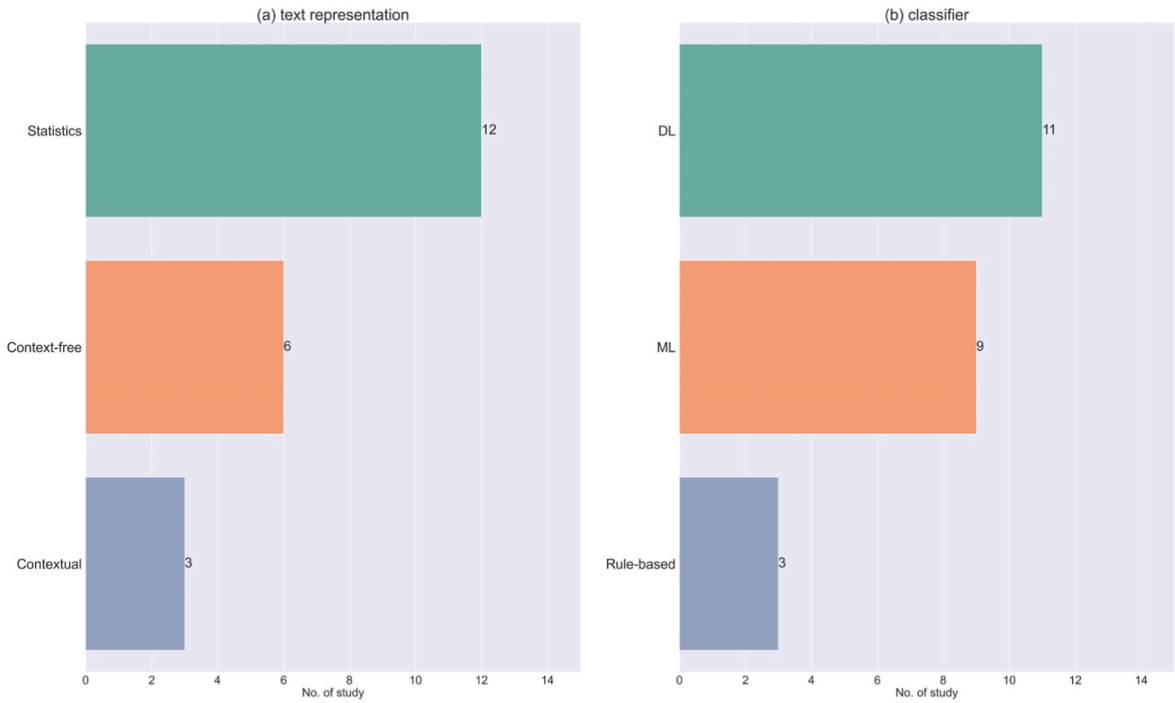

Fig. 2. Frequency of studies grouped by text representation (a) and classifier (b).

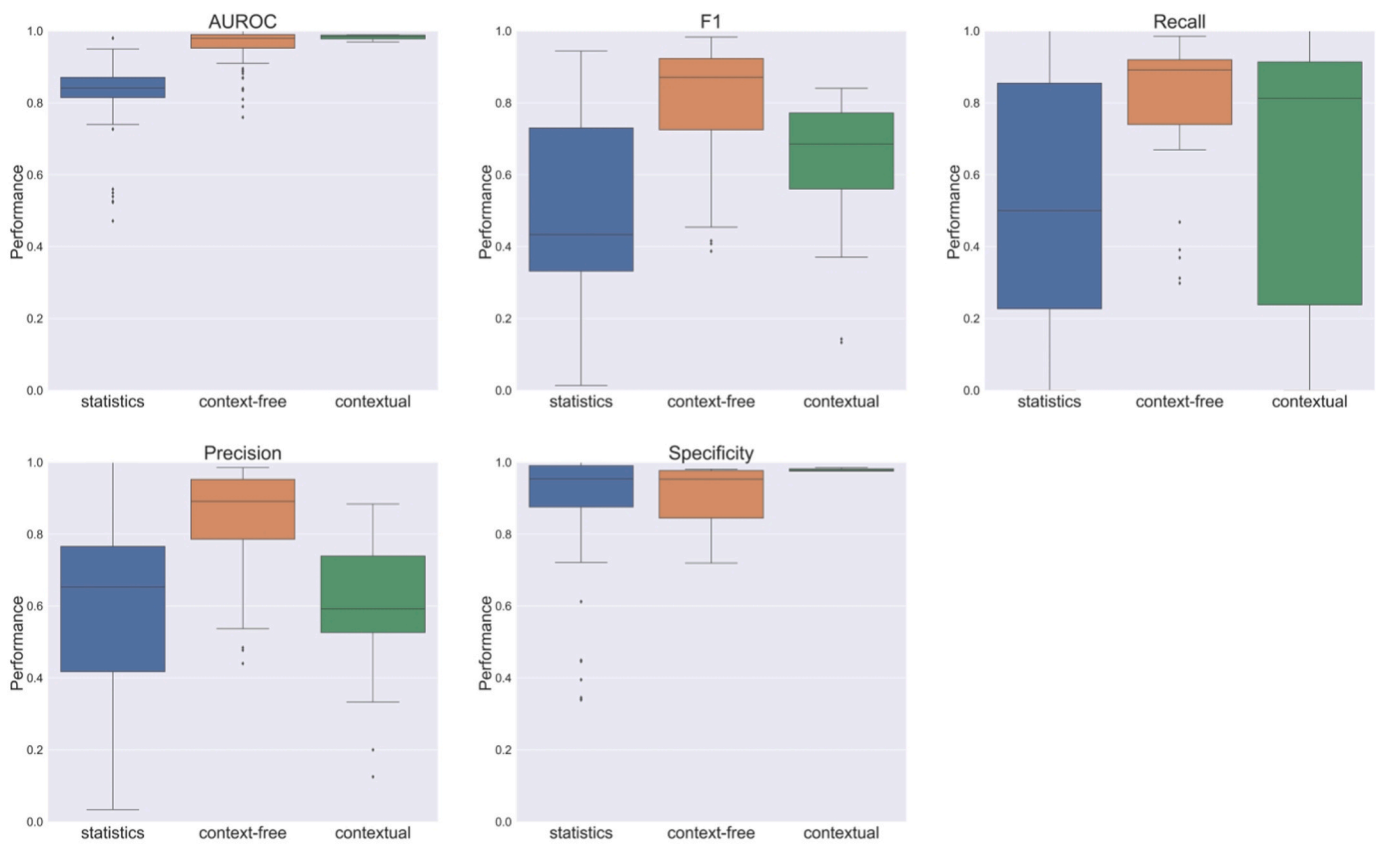

Fig. 3. Evaluation metrics of text representation techniques in the test data.

self-trained on neural word embedding from MIMIC-III data. For example, Wang et al. [38] applied CUI2vec, leading to improved model performance compared to TF-IDF approaches. Batch et al. [45] showed CNN with neural word embedding outperformed CNN with TF-IDF given consideration of the context of surrounding words within a sentence, in contrast to TF-IDF approaches which only consider the frequency of words. However, the improvement associated with word2vec has not always been consistent. This may be due to a limitation commonly

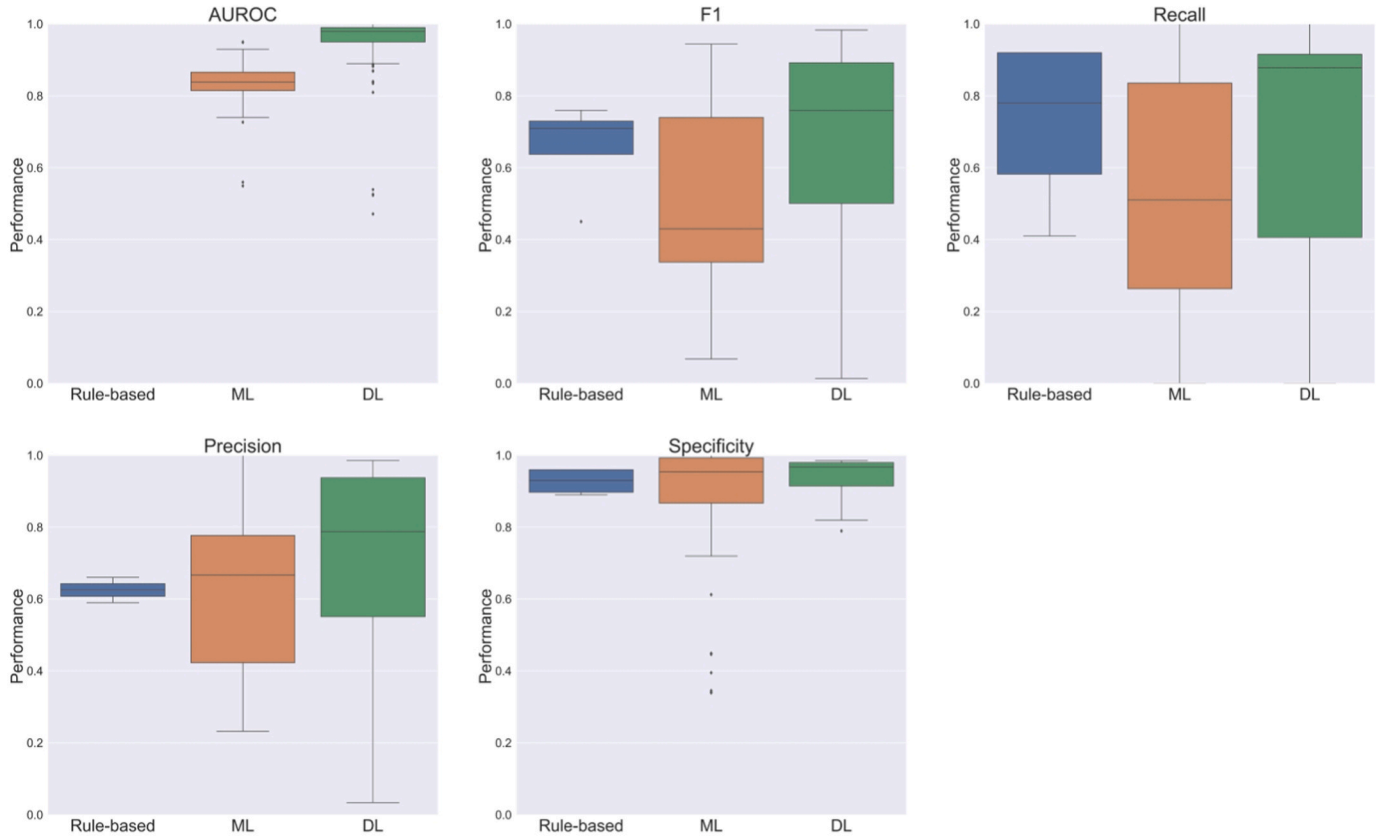

Fig. 4. Evaluation metrics of classifying models in the test data.

referred to as "out-of-vocabulary", where many clinical terms are not represented within the word embedding vocabulary used because word2vec was originally developed using Google News data, which more broadly represents general text terms rather than specific medical terminology. Moreover, context-free word embedding is further limited by being unable to differentiate words that may have different meanings, as such words would have the same representation. For instance, "pelvis" referring to the funnel area of the kidney would have the same embedding vector as "pelvis" referring to the articulation of the hip bones.

Only three studies [40,46,47] applied state-of-the-art NLP models at the time of review, applying BERT [22] and its variants [54–56], which consider contextual word embedding. Do et al. [40] applied BERT and TF-IDF to clinical-texts for metastatic classification yielding fair performance for both, with micro average scores of F_1, recall, and precision of 0.76, 0.78, and 0.71, respectively, for BERT and 0.76, 0.81, and 0.71 for TF-IDF. The low recall and precision observed may be due to the rarity of the metastatic organs affected (such as the kidney, spleen, and bowel), with sparse representation in the training data. Kaka et al. [46] applied BERT and ClinicalBioBERT [56] to clinical notes of breast cancers, achieving recall and specificity of 0.93 and 0.99 for BERT and 0.93 and 0.98 for ClinicalBioBERT. Although their models achieved a 95% reduction in manual review, their models may have been biased due to overfitting by data leakage. Lui et al. [47] developed MetBERT, a modified version of PubMedBERT, using MIMIC-III data with external validation using a local dataset that yielded moderate model performance (F_1, recall, and precision of 0.80, 0.77, and 0.82). However, their model was constrained by the maximum text input length of 512 tokens, with discharge summaries typically containing longer text entries. They used the first 512 tokens to develop their training model, discarding the remainder, which likely resulted in a loss of information.

A strength of BERT includes the use of a self-attention mechanism [22,29] that identifies relationships between all tokens within a document. However, BERT also has several limitations [57]. First, it is restricted to a maximum input length of 512 tokens. Other methods such as the Reformer [58], mitigate this limitation by managing text lengths as long as 4,096 tokens. The drawback, however, is that it applies word piece tokenization, which splits an original word into multiple subwords. For instance, "lymphoma" may be reduced to "l-ym-ph-oma", leading to potential misunderstanding of the original word. Applying health domain-specific pre-trained models, such as PubMedBERT [54], ClinicalBERT [56], and BioBERT [55], should help mitigate these concerns.

Among the three categories of algorithms, DL approaches performed best for the classification of recurrent cancer across all metrics, with median scores (IQR) of 0.98 (0.95–0.99) for AUROC, 0.88 (0.41–0.92) for recall, 0.79 (0.55–0.94) for precision, and 0.76 (0.50–0.89) for F_1. The most popular DL algorithm was CNN. Kehl et al. [35] used CNN to identify distant metastasis from radiology reports, achieving a median (IQR) F_1 score of 0.75 (0.71–0.83). Batch et al. [45] used the bidirectional LSTM algorithm for detecting cancer metastasis from radiology reports, reporting a median (IQR) F_1 score of 0.91 (0.87–0.97). RNN and its variants, are sequential models that are more likely to perform poorly when dealing with long text documents. This is because the models are unable to train in parallel, as the process of encoding is performed word by word from the beginning to the end of a document, and therefore prior loss of information may be problematic [59], reducing model performance. Attention mechanisms that simultaneously parallel process all words within a document mitigate this issue. Batch et al. [45] also included self-attention layers in CNN algorithms, which resulted in a high F_1 score with a median (IQR) of 0.98 (0.91–0.99), the highest performance relative to simple CNN and LSTM. BERT is a more complex model that attaches multi-head self-attention and applies only a linear layer for classification providing faster training times compared to bidirectional LSTM [54]. Nevertheless, BERT masks random tokens for prediction during the pre-training phase without masking during the

fine-tuning phase, which is known as a pretrain-finetune-discrepancy. BERT also assumes the predicted tokens are independent of other unmasked tokens, which is different to natural language, which has long-range dependency [60].

Even though our systematic review uniquely focuses on NLP techniques for recurrent cancer detection, we found that text representations and classifiers were similar to other medical domains. Seinan et al. [61] conducted a systematic review on the use of unstructured data for prognostic clinical prediction models. Their findings indicated that one-third of studies that applied bag-of-words and TF-IDF, with concept extraction and context-free word embedding represented in 14% and 18%, respectively. Another study focused on sepsis prediction from clinical text [62] also reporting comparable findings, with the majority of studies using TF-IDF and bag-of-words and only a single study applying contextual word embedding (Clinical BERT). Regarding classifier performance, our findings align with other systematic reviews. The common ML algorithms used were SVM and LR, while the common DL approaches included CNN, RNN variants [63,64] and linear classifiers of BERT. The linear classifiers of BERT performed best, followed by RNN variants [LSTM and gated recurrent unit (GRU)] [65].

In terms of reporting model performance, AUPRC likely represents the most informative metric relative to AUROC, as most datasets reflect imbalanced classes. Nevertheless, AUPRC was reported in only a single study [35], preventing comparison of this metric across studies.

Recently, new language models have been developed to address the limitations of BERT, including generative pre-trained transformer 3 (GPT-3) [66]. It was developed from 175 billion parameters by OpenAI in 2020 using a "few-shot learning" framework [66], which requires smaller labeled data for initiation compared to BERT. In addition, most ML approaches use supervised models, which require labeled data, limiting applications in healthcare domains because of the high cost and time-intensive nature of annotations by medical professionals; few-shot learning may mitigate these concerns. However, GPT-3 was originally trained on internet data which may introduce concerns related to social bias such as gender, racism, misinformation, and reliability [67]. In addition, it is not an open-source software, and the current costs of $120 and $30 per million tokens for direct-use and fine-tuning purposes, respectively, may prove prohibitive for some.

Similarly, sentence embedding approaches such as sentence BERT (SBERT) [68], unlike word embedding, convert an entire sentence into a vector representation known as sentence representation. Gatto et al. applied SBERT for COVID-19 severity classification from online health platforms [69] reporting improved SBERT performance compared to conventional BERT and BioClinicalBERT. Future research should consider the use of domain-specific language models for biomedical terms, such as PubMedBERT [54], CancerBERT [70], and BioGPT [71]. Additionally, developing automated cancer detection tools for specific cancer types using NLP techniques, including contextual word embedding and DL algorithms, could prove beneficial.

To the best of our knowledge, this study is the first and only systematic review to evaluate NLP techniques, including text representation and algorithms for cancer recurrence detection. This study followed standard guidelines for conducting systematic reviews, including adhering to PRISMA2020 and TRIPOD-SRMA guidelines. The findings from our study may inform the development of an automated cancer recurrence detection tool. However, our systematic review had several limitations. First, our review only included relevant studies from the four database searches, so conference proceedings or full studies published in other gray databases may have been overlooked. Second, we did not apply meta-analysis to pool algorithm performance, given the variable reporting of model performance metrics. Third, this field is still relatively novel, and head-to-head comparisons of different methods are still lacking.

5. Conclusion

This systematic review summarizes the evidence to determine the best model and text representation technique for recurrent cancer classification from EMRs. The findings suggest that DL models with text representation of PubMedBERT trained from medical domains provide the best performance for clinical text fields. Moreover, this review will inform clinicians, researchers, and developers with regards to the most appropriate approaches for recurrent cancer detection from EMRs.

Funding

This study was funded by the National Research Council of Thailand (N42A640323), Thailand. The funding agency had no role in the design or conduct of the study.

Author Contributions

E.S.: Methodology, study selection, Data Curation and extraction, Formal analysis, Writing - Original Draft. **W.P.**: Methodology, study selection, Data Curation and extraction, Formal analysis, Writing - Review & Editing, Supervision. **A.Ta.**: Methodology, study selection, Data Curation and extraction, Writing - Review & Editing, Supervision. **N.T.**: Conceptualization, Methodology, Data Curation and extraction, Formal analysis, Writing - Review & Editing, Supervision. **P.N.**: Conceptualization, Supervision, **G.J.M.**: Writing - Review & Editing, **J.A.**: Writing - Review & Editing. **A.T.**: Conceptualization, Methodology, Writing - Review & Editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This manuscript is a part of Ekapob Sangariyavanich training in international Ph.D. program (Data Science for Healthcare), at the Department of Clinical Epidemiology and Biostatistics, Faculty of Medicine Ramathibodi Hospital, Mahidol University, Bangkok (CEB), Thailand. This study is a part of his dissertation.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.imu.2023.101326.

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